

Cycle vs event rebuild time for three planners

A typical rebuild is far below the 100 ms limit (95 % of rebuilds finish within 12–34 ms), and rebuilds over 100 ms are rare (at most 68 per 10,000). Each rebuild is a new chance of one, so rebuilding less often stops fewer robots: at 20 ms checks, cycle vs event Voronoi 17 vs 2, PRM 27 vs 10, Visibility Graph 52 vs 27. Rebuild time is mostly map construction; the Dijkstra search is a few percent. Times are wall clock on one machine, successful rebuilds only. ▲ blue (bold) = better of the pair, ▼ orange = worse; times marked only when they differ by 10 % or more.

	Rebuilds per run	Time per rebuild, median (ms)	95 % of rebuilds under (ms)	99 % of rebuilds under (ms)	Slowest rebuild (ms)	Rebuilds over 100 ms per 10,000	Share of time in Dijkstra search (%)	Robots stopped by 100 ms limit
Voronoi								
Cycle @ 20 ms	88.2 ▼	5.9 ▼	12.8	28.1	556 ▼	10	1	17 ▼
Event (ours) @ 20 ms	10.8 ▲	4.5 ▲	12.2	26.8	107 ▲	9	2	2 ▲
Cycle @ 100 ms	16.9 ▼	5.8 ▼	12.2	22.1 ▲	108 ▲	3 ▲	1	1
Event (ours) @ 100 ms	6.1 ▲	4.7 ▲	12.3	33.6 ▼	471 ▼	8 ▼	1	1
PRM								
Cycle @ 20 ms	70.4 ▼	11.7	29.0	50.7 ▲	1420 ▼	19 ▲	1	27 ▼
Event (ours) @ 20 ms	7.4 ▲	11.8	29.2	67.7 ▼	233 ▲	68 ▼	1	10 ▲
Cycle @ 100 ms	13.6 ▼	11.9	29.6	54.5	456 ▼	29 ▲	1	8 ▼
Event (ours) @ 100 ms	4.4 ▲	12.5	29.3	56.5	275 ▲	57 ▼	1	5 ▲
Visibility Graph								
Cycle @ 20 ms	94.6 ▼	14.8 ▲	26.7 ▲	54.5 ▲	1879 ▼	27 ▲	3	52 ▼
Event (ours) @ 20 ms	20.8 ▲	16.3 ▼	34.0 ▼	78.1 ▼	272 ▲	65 ▼	3	27 ▲
Cycle @ 100 ms	18.5 ▼	15.1	30.1	81.1	610 ▼	57 ▼	3	21 ▼
Event (ours) @ 100 ms	8.1 ▲	16.6	32.5	80.7	188 ▲	50 ▲	3	8 ▲